Modeling light pollution using geospatial, meteorologoical, energy usage, and population data

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DSA 5103– Fall 2025

December 9, 2025

Contents

List of Figures

List of Tables

1 Executive Summary

Problem Statement Light pollution is a night-time phenomenon caused by the atmospheric scattering of excess light from things such as street lamps or other outdoor light fixtures that then bounces off of molecules in the air. Light pollution is commonly measured as night sky brightness (in magnitudes per square arcsecond), but is typically only directly measured at sites like observatories. Even then, data from these observatories is rarely accessible in real-time. The Globe at Night campaign created a monitoring network for observatories across the world to provide night sky brightness data for public use, with data going back as far as 2014. Light pollution, being an atmospheric phenomenon, is heavily impacted by weather conditions. The objective for this project is to predict light pollution (night sky brightness) at a given location using primarily meteorological, energy usage, and population data.

Major Concerns/Assumptions The training data for night sky brightness is limited to thirteen observation sites with relatively sparse readings is expected to be a significant training obstacle, likely to skew the models locally to those geographic locations. Additionally, the association of zip-codes to the nearest geographic electric-service provider is a rough approximation that may not accurately reflect the actual service-provider for a given area, most notably for high population density zones that are serviced by multiple providers. Population data is also limited to the static values collected during the 2020 US Census. Time-based population estimation methods were not used. Ultimately, the site data for 'SR' Silver Ranch in TX was dropped from consideration due to abnormal measurement distributions on the high-end of night sky brightness compared to the rest of the sites.

Summary of Findings The final trained MARS model made predictions against a 25% data test-holdout set with a 5-repeated 10-fold CV performance of $RMSE = 0.6731$, $R^2 = 0.9093$, and $MAE = 0.4209$. Concerningly, the final model pruned nearly all meteorological variables, placing emphasis nearly entirely on relative sun/moon positions, time of year/day, and locality to population/electrical centers.

Recommendations The model predictions must be validated against an additional site with different local meteorological conditions. If the model stands up to generalization, then further refinement can be done focusing within tuning the model's predictions on the low-end of night sky brightness where the most outliers are present.

2 Problem Background

2.1 Problem Description

The use of artificial light at night (ALAN) in outdoor spaces—e.g., through the use of light fixtures such as street lamps or flood lights—causes what is known as light pollution. More specifically, improper and excessive use of ALAN, which has risen dramatically alongside rapid urbanization and industrialization around the globe, is the primary source of light pollution. The definition of light pollution varies across different agencies, but the common denominator amongst them all is that it is the culmination of the unwanted, adverse aspects of ALAN. There are three main components that make up light pollution: glare, light trespass, and skyglow; for the purposes of this project, however, the focus is solely on skyglow[7]. Skyglow, the most broadly impactful aspect of light pollution, is the result of light being emitted upwards and then bouncing off of particles in the Earth's lower atmosphere, scattering. It obscures the natural night sky, preventing those in the city from being able to marvel at the stars. Those with exceptional vision and access to pristine night skies are able to see up to 7,000 stars with the naked eye, however, that number plummets to fewer than 100 for the same observer in an urban area[5].

As it currently stands, light pollution-free skies are becoming more and more difficult to find. Over 99% of US and European populations are affected by some level of light pollution, and a recent analysis found that skyglow is rapidly worsening at a rate of nearly 10% per year[4]. Modeling the skyglow component of light pollution, typically referred to as night sky brightness, is a crucial aspect of many light pollution studies. Unfortunately, there is no definitive way to assess skyglow, and each method has its own pros and cons. Light pollution data for modeling often comes from a combination of ground-based measurements and satellite data. Ground-based measurements are most often taken using a sky quality meter (SQM), but have also been derived from citizen-science reports of naked-eye limiting magnitude (NELM). NELM is the magnitude (a logarithmic scale describing the apparent brightness of a star) of the faintest visible star to a naked-eye observer; the lower the magnitude, the brighter the object. The SQM is a handheld sensor that measures the brightness of the sky directly overhead in units of magnitudes per square arcsecond.

Due to the complexity of light emission behavior and limitations in computing power, research into different modeling methods is still ongoing [3]. Most light pollution models are numerical models based on a series of functions that describe upward light flux and atmospheric scattering. Many more radiative transfer-based numerical models exist, but are often tailored towards modeling meteorological conditions and therefore not suitable for modeling light pollution specifically [3]. Recently, it was found that most, if not all light pollution models are significantly, systematically biased through implicitly assuming that all particles in the lower atmosphere are homogenous and spherical (Mie theory), despite being a safe assumption for daytime light models [2]. A popular global map known as the New world atlas of artificial night sky brightness [1] is another numerical model utilizing both ground-based and satellite measurements, but assumed a perfectly clear, calm, atmosphere over the entire globe. This limitation, albeit understandable, greatly impacts the accuracy of its output. Atmospheric conditions vary widely across regions. Not only that, but atmospheric conditions are always changing as well. Consecutive measurements of skyglow in the exact same location, even just hours apart, could produce different results.

While acknowledging that the extent of light pollution is influenced by a multivariate set of economic indicators [6], the current model specification is limited to assessing the predictive power of population metrics and energy consumption data.

2.2 Data Description

Night-Sky Brightness Globe at Night (GaN) The GaN-MN project, an extension of the original Globe at Night project, is a global night sky brightness monitoring network using a commercially available meter (SQM-LE by Unihedron) for long-term monitoring of the light pollution conditions in different places around the world.
https://globeatnight.org/gan-mn/

Meteorological Data ERA5 provides hourly estimates for a large number of atmospheric, ocean-wave and land-surface quantities. An uncertainty estimate is sampled by an underlying 10-member ensemble at three-hourly intervals. Ensemble mean and spread have been pre-computed for convenience. Such uncertainty estimates are closely related to the information content of the available observing system which has evolved considerably over time. They also indicate flow-dependent sensitive areas. To facilitate many climate applications, monthly-mean averages have been pre-calculated too, though monthly means are not available for the ensemble mean and spread.
https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview

Zip-Code Populations A tabulation of the total population per zip code per the 2020 U.S. Census.
https://data.census.gov/table

Electric Retail Service Territories A table of electric power retail service territories in the U.S. These are areas serviced by electric power utilities responsible for the retail sale of electric power to local customers, whether residential, industrial, or commercial.
https://ornl.opendatasoft.com/explore/dataset/electric-retail-service-territories/information/

Population-Weighted Centroids for Zip Code Tabulation Areas (ZCTAs) This dataset denotes ZIP Code centroid locations weighted by population.
https://hudgis-hud.opendata.arcgis.com/datasets/d032efff520b4bf0aa620a54a477c70e/about

2.3 Exploratory Data Analysis

Night-Sky Brightness The Globe at Night dataset only contains thirteen sites and night sky brightness readings are relatively sparse (∼63% missing); however most of the missing measurements are from the daylight hours. Additionally the minimum time resolution of measurements is to the UTC hour instead of the local hour. A box-plot of the NSB measured at each site is shown in Figure 1.

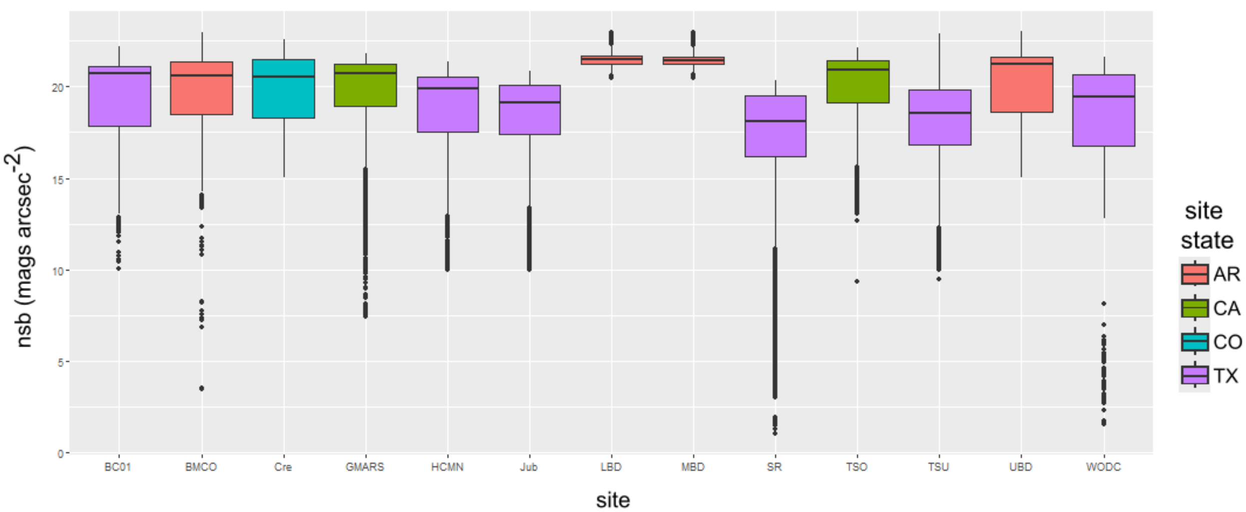


Figure 1: Box-plot of Night Sky Brightness Measurements by Site

Meteorological Data Given that ERA5 is a reanalysis dataset, there are no missing values that must be accommodated.

Electric-Service Provider Coverage The service territories dataset is missing zip-code data for approximately 58% of electrical service providers.

Population 4320 zip-codes are missing between the intersection of the centroids and population datasets. This is due to zip-codes that did not have population data in the 2020 census data.

Principal Component Analysis Omitting the night-sky brightness (NSB) from the variables compiled via the sources above (26 total), when performing a PCA 43.4% of the data-set variance is contained within components 1 and 2. A random sample of 5% of the data was selected to plot against PC1 and PC2 for easier viewing, with each point colored by a gradient for the associated NSB measured value. This bi-plot is shown in Figure 2. A cursory inspection of the plot implies there may be a skew of low NSB nights (the points of interest) in the direction of PC1.

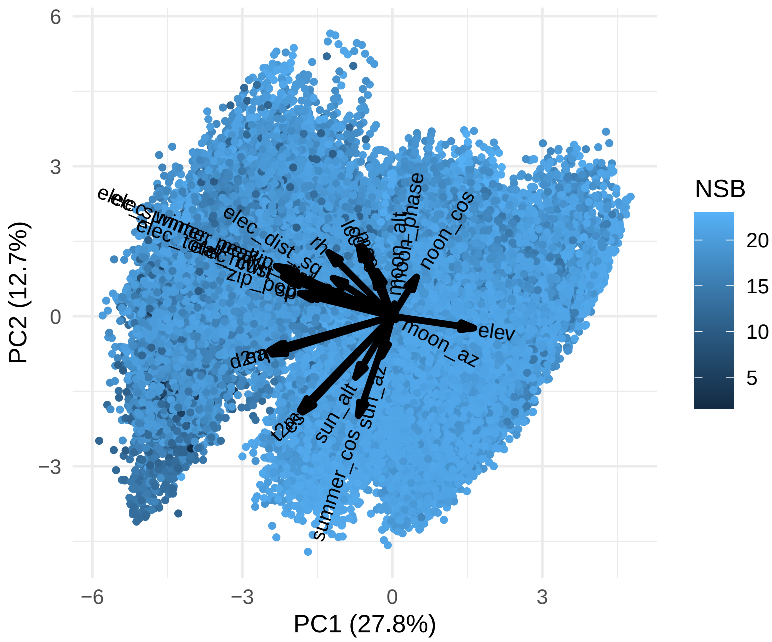


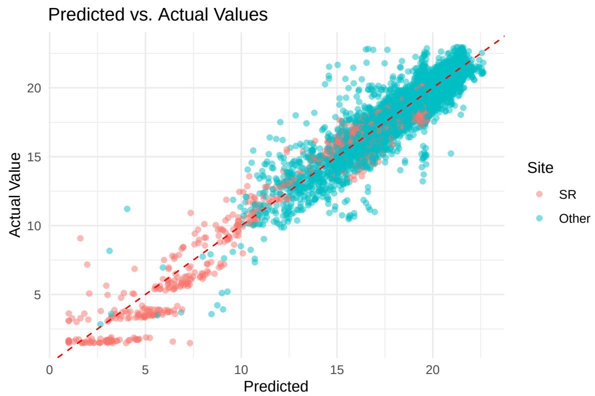
Figure 2: PCA Bi-Plot w/ Night-Sky Brightness Gradient

3 Methodology

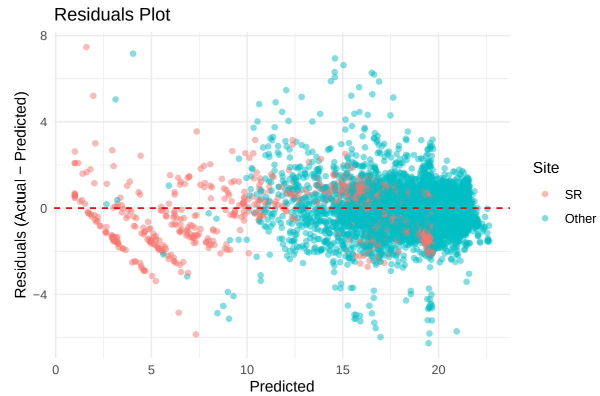
3.1 Feature Selection

Night-Sky Brightness Missing brightness readings are dropped from the training data entirely. The time of observation is assumed to be close enough to local time, so no time zone adjustments are made. The brightness readings are assumed to be correlated to the time as a cyclical feature, so the month and hour of observation are encoded as wave functions at an angular offset from typical brightness zeniths. Night sky brightness observations are recorded approximately every 5 minutes. However, the meteorological data is provided at an hourly resolution, so the mean night sky brightness of each hour is calculated and used instead. A local timestamp, as well as a UTC timestamp is recorded at the time of each observation as well, but only the UTC timestamp was kept, as the ERA5 timestamps are provided in UTC.

NOTE: After a preliminary model evaluation, measurements from the `SR` site were creating strange residual behavior in the model. Because this site was unique in the amount of bright sky measurements and had anomalous patterns in its readings compared to the rest of the sites, it was removed from the training and testing data. A comparison of the SR site measurements and the rest of the sites are shown in Figure 3



(a) Predicted v. Actual



(b) Predicted v. Residual

Figure 3: 'SR' Site vs Others

Meteorological Data ERA5 is currently one of the most popular and comprehensive meteorological reanalysis datasets publicly available, with a horizontal spatial resolution of 0.25 by 0.25 degrees and an hourly temporal resolution. While ERA5 provides a very large variety of atmospheric/meteorological variables, only the variables air temperature, dewpoint temperature, surface air pressure, and percentage of high cloud cover, medium cloud cover, and low cloud cover were chosen for use. With the python package `astropy`, using a site's location and time of observation, the altitude and azimuth of the sun and moon were calculated to serve as feature variables as well.

Electric-Service Provider Rows with missing zip-code data for service providers are dropped from the training data due to the inability to map given providers to sensible zip-codes.

Population The zip-code centroids with missing population data are dropped from the training data. The values for population are not being estimated for time-adjusted values.

Expected Energy Usage The assumption is made that energy usage by nearby population centers is correlated to their nearest electric-service providers. Using the population-centroids of each zip-code, the nearest electric-service provider is associated with each population center. The nominal energy usage of each population center is then encoded as a sine function of the month of observation to differentiate between the seasonal energy usage patterns. The `SUMMR_PEAK` and `WINTR_PEAK` values are used as the amplitudes of the wave functions for the respective seasons.

Distance from Population Centers Using the Haversine formula, the great circle distance between the nearest zip-code population centroid and the observation site is calculated and included as a feature.

Since the radiance from light-sources are known to decay with distance, this distance is encoded as its inverse square.

3.2 Model Choice

The `earth` R package was used for its implementation of the MARS regression model. The `caret` package was used to train and tune the hyper-parameters `nprune` and `degree` of the model. Post-processing will be applied before the final prediction to clamp negative values up to the minimum magnitude of 1.

3.3 Validation Approach

The training control used for the model training was 10-fold CV with 5 repeats. The full dataset was split 75/25 between the training and testing holdout data.

4 Results

4.1 Model Results and Performance

Training Results The model training results are shown in Table 1. The hyper-parameters were tuned up to `degree`$= 5$ and `nprune`$= 45$, with marginal improvements in $RMSE$ with each expansion of the tuning grid after `degree`$= 4$ and `nprune` > 30. More fine tuning of the hyper-parameters is possible, but the performance appears to be relatively plateaued with the current predictors.

`degree`	`nprune`	**RMSE**	**R^2**	**MAE**
2	25	0.8809	0.8404	0.5562
2	30	0.8701	0.8443	0.5482
2	35	0.8691	0.8447	0.5479
2	40	0.8691	0.8447	0.5479
2	45	0.8691	0.8447	0.5479
3	25	0.7640	0.8799	0.4749
3	30	0.7419	0.8864	0.4590
3	35	0.7224	0.8923	0.4465
3	40	0.7160	0.8940	0.4394
3	45	0.7118	0.8952	0.4364
4	25	0.7644	0.8794	0.4707
4	30	0.7326	0.8893	0.4547
4	35	0.7124	0.8953	0.4419
4	40	0.6951	0.9002	0.4296
4	45	0.6822	0.9039	0.4192
5	25	0.7638	0.8796	0.4679
5	30	0.7360	0.8882	0.4525
5	35	0.7140	0.8948	0.4400
5	40	0.6980	0.8992	0.4283
5	45	0.6863	0.9025	0.4190

Table 1: Summary of `caret` training results for `earth`

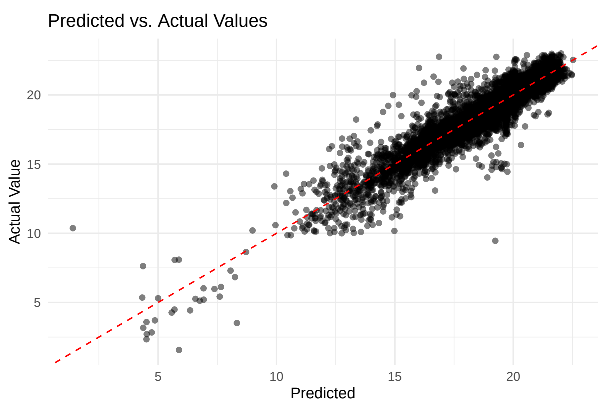
Post-Processing After clamping predicted magnitudes < 1 up to 1, the resampled performance yielded marginal improvements across the categories as shown in Table 2.

Residual Analysis Figure 4a shows a plot demonstrating the post-processed prediction of the NSB against the actual NSB measured magnitude. Figure 4b shows a plot demonstrating the residual values for each predicted NSB magnitude. From these plots it is seen that the model has been relatively strongly

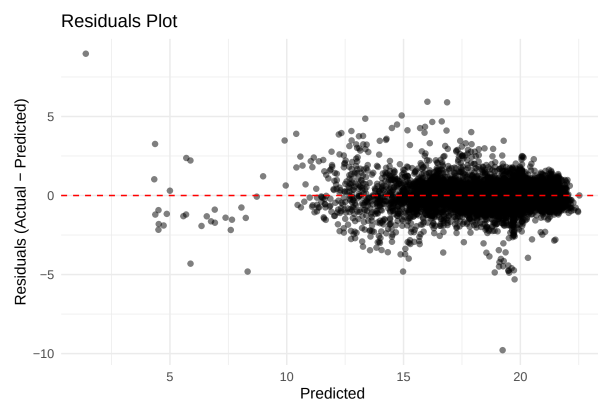
RMSE	R^2	MAE
0.6731	0.9093	0.4209

Table 2: Performance of Post-Processed Predictions

impacted by the large cluster of dark nights in the dataset. The residuals are roughly evenly distributed between the predictions, and even the outliers in the bright/dark ends of the results are symmetric with respect to each other.



(a) Predicted v. Actual



(b) Predicted v. Residual

Figure 4: Predicted Values and Residuals

Variable Importance After the relatively aggressive pruning parameter of `nprune`= 45 in the final model, the original set of 26 predictors were trimmed down to a 4-way interaction between the 14 variables listed in Table 3.

4.2 Key Findings

Final Variables The intercept of the final model starts at 19.7, meaning the model starts with expecting a dark sky. Inspecting the hinge functions of the final model show that the largest impacts on NSB were sun altitude, nearby population, distance to nearby population centroid, distance to nearest electrical service provider, proximity to noon, and cloud cover in the top 8 splines. The moon altitude and moon phase begin to appear in all interactions after this point with the other variables. Interestingly most of the weather variables were pruned away.

However, because of the initial limitation of the dataset to the thirteen sites that participate in the GaN project in the USA; we suspect it is likely due to the weather effects specific to these sites being averaged

variable	**nsubsets**	**gcv**	**rss**
moon_alt	44	100.0	100.0
sun_alt	43	79.3	79.3
noon_cos	42	67.6	67.6
elev	41	57.2	57.3
zip_pop	40	51.2	51.2
moon_phase	39	44.6	44.6
t2m	36	32.3	32.3
sun_az	33	27.7	27.7
elec_dist_sq	33	27.7	27.7
summer_cos	30	25.3	25.4
cc	28	22.6	22.6
zip_dist_sq	21	15.5	15.6
elec_winter_peak	15	12.5	12.5
lcc	11	10.3	10.4

Table 3: Final `earth` Model Variable Importance

away in the tuned coefficients. Unfortunately, to verify this would require purchasing the SQM-LE research tool to measure the night sky from off-nominal locations. We believe that the model may be over-tuned to the specific environments of the thirteen sites.

5 Conclusion

Summary We set out to predict the night sky brightness for a given location based on detailed local weather data, proximity to nearby population centers, local electricity usage, and time of day/year. Using supervised learning on the combined datasets between GaN, ERA5, US-DOE, and the US Census Bureau; a MARS model was trained to perform regression analysis through the `earth` R-package to yield a final model with predictive night-sky brightness magnitudes with `RMSE`= 0.6731. Combined with an $\mathbf{R}^2 = 0.9093$, the model is either over-tuned for the thirteen site restriction of the dataset, or it is a fantastically accurate model.

Key Insights It is important to reiterate that the training dataset for the night sky brightness was limited to the US based sites that report the data. The distances, populations, and electrical information are static with only thirteen possible values corresponding to their respective sites. Whether or not this extends generally to any location remains to be seen.

With this restriction in mind, We find that the model pruned away all variables except for ones relating to the season (e.g. `summer_cos` is a feature engineered as the cosine of the month of a measure relative to June) or time of day (relative to noon), sun/moon position, nearby population and distance, and local energy usage. Somewhat mysteriously, the only weather predictors that survived the pruning was the `t2m` temperature and both cloud-cover variables.

The primary suspect is that the average of the weather effects across the years between all thirteen sites was baked into the model hinge function coefficients.

Final Recommendations and Impact While acknowledging the constraints of the training data, the model's performance on the available set is promising. If the model's generalization error is found to be close to its training performance, it offers significant practical utility:

- **Impact:** The model provides hobbyist stargazers with a tool to plan for optimal conditions and offers professional astronomers a data-driven aid in the selection or evaluation of potential new observatory sites.
- **Caution:** The highly localized nature of the training data necessitates caution. The strong correlation with static site characteristics (2020 Census populations, local energy providers) means the model's high R^2 may not generalize well to novel locations outside the established parameters of the thirteen training sites. Further validation on independent site data is imperative to determine the model's true scope and utility.

References

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